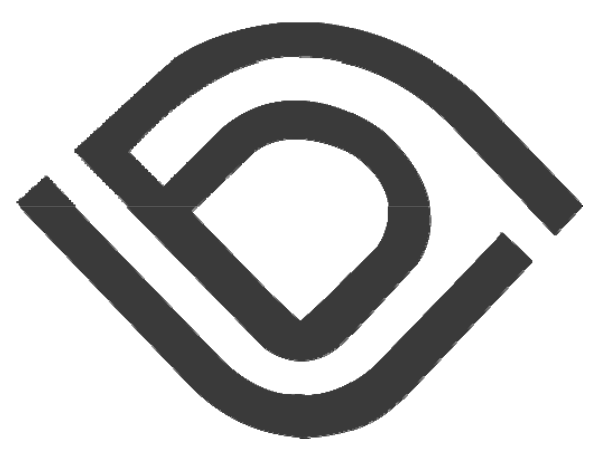




Data Visualization Lab
Sejong University

Scanpath Visualization According to Fixations with Gazed Objects

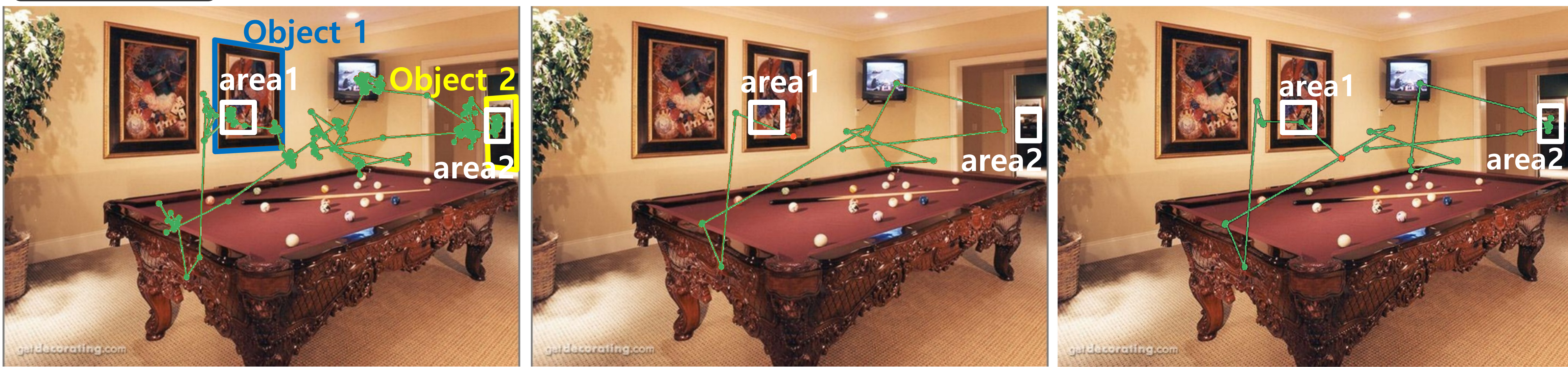
Sangbong Yoo⁰, Soobin Yim, Chanyoung Yoon, Hyein Hong, and Yun Jang[†]

Sejong University



We propose **an object and velocity-threshold based fixation identification algorithm** for **integrating semantical information** with eye movements.

Motivation



(a) Raw gaze points

(b) Fixations missing with I-VT

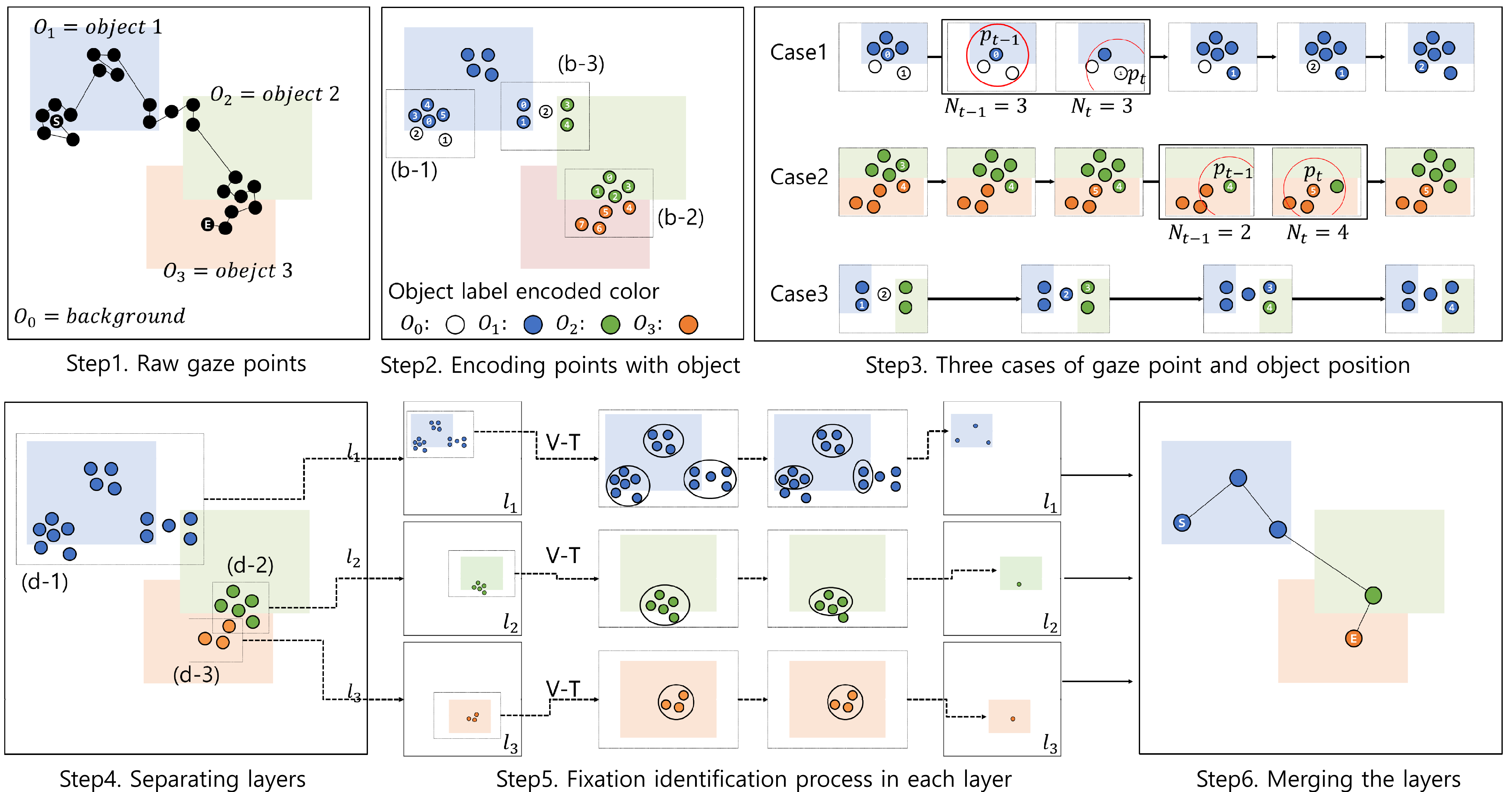
(c) Scanpath with our algorithm

- The **interpretation** of the observer's **gaze behavior depends on** the fixation location.
- In (b), the I-VT **misses fixations** in area 1 and area 2.
 - Existing algorithms **do not use objects** for fixation identification.
- In (c), our proposed algorithm **identifies fixations with segmented objects** in area 1 and 2.

The proposed algorithm **interprets the visual search without missing the objects gazed at** by the observers.

Our algorithm for object and velocity-threshold fixation identification

Our algorithm identifies **fixations** with **segmented objects**, eye movement data, **velocity threshold**, **foveal area range**, and time.



Step4. Separating layers

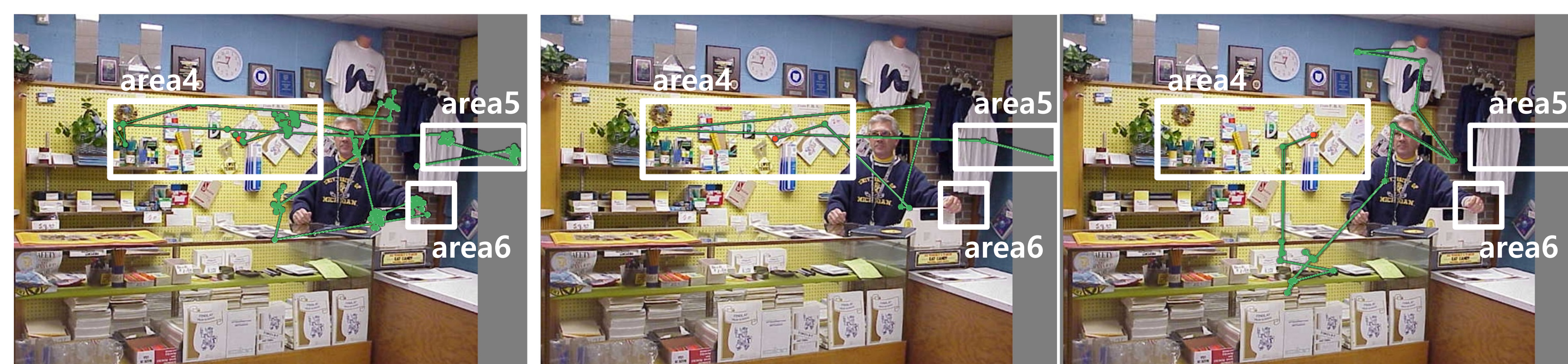
Step5. Fixation identification process in each layer

Step6. Merging the layers

Comparisons



- In scanpath with I-VT, the fixations move from **object 3** to **object 5** without going through **object 4** as fixation is not identified as in **area 3** although there are data.
- The scanpath with our algorithm **shows the fixation** in **area 3**.
- Ours **do not miss** the **visual searching process** of observers.



- However, our algorithm has limitations.
 - Ours has **poor performance** for visual stimulus with **many small objects**.
 - It is **dependent on segmented object** dataset.
 - Also, segmented objects **manually labeled** by humans are **subjective**.

Raw gaze points

Scanpath with I-VT

Our algorithm